DTU Compute
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Machine Learning for Semi-Automated Annotations of the Linear A Inscriptions

Bachelor Thesis

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Preface

This Bachelor's thesis was prepared at the Department of Applied Mathematics and Computer Science at the Technical University of Denmark in fulfillment of the requirements for acquiring a Bachelor of Science in Engineering (BSc Eng) degree in General Engineering (Cyber Systems).

Kongens Lyngby, March 23, 2026

Frederik W. L. Christoffersen

Abstract

This thesis is in the workings... [Write something when finished with all thesis chapters]

Acknowledgements

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Generative AI tools were occasionally used as support for brainstorming, linguistic editing, code exploration, and research during the development of this project.

Contents

Preface **i**

Abstract **iii**

Acknowledgements **v**

Contents **vii**

Acronyms **ix**

List of Figures **x**

List of Tables **xii**

1 Introduction **1**
1.1 Background 1
1.2 Prior work 5
1.3 Problem statement 6
1.4 Empirical considerations 7
1.5 Impact – innovation and application 8
1.6 Overview of the thesis 8

2 Data **9**
2.1 Sources 9
2.2 Limitations and biases 11

3 Methodology **13**
3.1 Common methodology 13
3.2 Segmentation quality 14

3.3 Interactive semi-automation 24
3.4 Workflow efficiency 24

4 Results **25**
4.1 Segmentation performance 25
4.2 Automation evaluation 27
4.3 Workflow efficiency evaluation 27

5 Discussion **29**
5.1 Segmentation ability 29
5.2 Automation and interactivity 29
5.3 Workflow impact 29

6 Conclusion **31**
6.1 Summary of findings per research question 31
6.2 Overall contribution 31
6.3 Final remarks 31

7 Future Work **33**
7.1 Methodological extensions 33
7.2 Dataset expansion and diversification 33
7.3 Further automation opportunities 33

Bibliography **35**

A An Appendix **39**
A.1 Raw segmentation performance 39

Acronyms

HT Haghia Triada Tablet

SAM Segment Anything Model

mSAM MobileSAMv2

mSAM+pts MobileSAMv2 with automatic point prompts

GC+brush GrabCut with brush seeding

Cefael Collections de l'École française d'Athènes en ligne

GORILA Godart and Olivier, Recueil des inscriptions en linéaire A

PDR Project Definition Report

IPR Intellectual Property Rights

GMM Gaussian Mixture Model

IoU Intersection over Union

Dice Dice-Sørensen coefficient

TP True Positives

TN True Negatives

FP False Positives

FN False Negatives

List of Figures

2.1 Example of a raw inscription image and its corresponding ground truth annotation. 10

2.2 Criteria for classifying the dataset into easy, medium, and hard difficulty levels. 11

2.3 Examples of images with different levels of segmentation difficulty: easy, medium, and hard. For each case, the raw image (left) and the corresponding ground truth annotation (right) are shown. 12

3.1 A visualization of seed distribution for GrabCut with brush seeding (GC+brush) on HT118. Green overlay represents sure foreground pixels, while red overlay represents sure background pixels, and absence of overlay represents probable background pixels. 20

3.2 A visualization of seed distribution for MobileSAMv2 with automatic point prompts (mSAM+pts), overlayed on HT118. Green points represent foreground seeds, while red points represent background seeds. 22

4.1 Comparison of segmentation outputs from different models against the raw image and ground truth. 26

4.2 Dice Score performance per model. Split by difficulty. With error bars representing the standard error of the mean. Ordered descendingly from the left by mean performance. . . . 27

4.3 QQplot of the residuals of the paired t-test comparing G+brush and mSAM+pts. The residuals appear to be approximately normally distributed, which supports the validity of the paired t-test results. Furthermore, the Shapiro-Wilk test for normality on these residuals yields a p-value of $5.9 \cdot 10^{-1} > \alpha = 0.05$, indicating that we fail to reject the null hypothesis of normality. 28

List of Tables

A.1 Raw Dice score data on easy images. Label column represents the name of the document segmented, while all other columns represent Dice scores obtained for those documents for a given model. 39
A.2 Raw Dice score data on medium images. Label column represents the name of the document segmented, while all other columns represent Dice scores obtained for those documents for a given model. 40
A.3 Raw Dice score data on hard images. Label column represents the name of the document segmented, while all other columns represent Dice scores obtained for those documents for a given model. 40
A.4 Summary Statistics per Model of Dice Score on all images. . 41
A.5 Summary Statistics per Model of Dice Score on easy images. 41
A.6 Summary Statistics per Model of Dice Score on medium images. 41
A.7 Summary Statistics per Model of Dice Score on hard images. 42

CHAPTER 1

Introduction

1.1 Background

When the British archaeologist Arthur Evans in the beginning of the 20th century went to excavate Knossos on Crete, he found a huge amount of clay tablets written in a previously unknown linear script. Some documents were clearly of a newer script, which he called *Linear B*, while others were of an older one, which he named *Linear A*. Some fifty years later, the British amateur archaeologist Michael Ventris finally succeeded in deciphering the newer script Linear B, which he proved to be representing Mycenaean Greek, the earliest attested form of Greek. But the same hypothesis did not hold for the older script Linear A, whose encoded language remained unknown, still yet to be deciphered.

A key obstacle to its decipherment has been the small size of its corpus, lately consisting of only around 2,500 inscriptions (about 1,400 if only including those well-preserved enough to be readable) - significantly smaller than that of its younger counterpart that now spans more than 4,500 inscriptions (which are comparatively longer and better preserved).[1,2] But new inscriptions are still being discovered, with the corpus most recently extended in 2025 by over 100 new documents.[3,4]

Another challenge has been the lack of digital representations of these inscriptions, in a form suitable for statistical analysis. While photographs with accompanying drawings of nearly the entire corpus are available online through Collections de l'École française d'Athènes en ligne (Cefael) (see Godart and Olivier (1976–1985)), these are only available in print

[1] Salgarella 2020.
[2] Salgarella 2025, p. 13, 46.
[3] Del Freo and Zurbach 2025.
[4] Consiglio Nazionale delle Ricerche 2025.

form. This motivated the development of the palæographical[5] database *The Signs of Linear A* (SigLA), which seeks to ”[develop] a systematic, exhaustive and user-friendly open access database of all Linear A inscriptions [...] in order to carry out statistical and palæographic analyses within the epigraphic corpus” (Salgarella and Castellan 2020). However, the last addition to SigLA was in 2021, and the project has so far managed to document just 772 of all currently known inscriptions.[6,7]

1.1.1 A problem of manual digitization

The drawings currently stored on SigLA were made by Ester Salgarella using the painting tool Krita. For each document, she manually created her own annotated drawings based on the originals from Cefael. From these, she proceeded to create layered Krita files with metadata for each sign, which could then at last be imported to SigLA.[8] According to her (personal communication), this has been a highly time-consuming undertaking. And the necessity of this kind of labor-intensive work has arguably been the biggest reason to why SigLA still does not contain the entire available corpus.

The aim of this project is therefore to investigate whether parts of this process can be automated. More specifically, whether some kind of semi-automatic segmentation tool can accelerate the workflow, while maintaining high-quality outputs that are still compatible with SigLA's database.

1.1.2 A problem of small data

The small size of the Linear A corpus and the irregularities in quality of its documents is a huge problem for the application of deep learning approaches, which typically require large amounts of data to perform well. Consequently, this project will instead seek to make use of generalized

[5]The term ”palæographical” refers to the study of ancient writing systems.
[6]SigLA 2021a.
[7]SigLA 2021b.
[8]Salgarella and Castellan 2020.

machine learning and computer vision, in particular Meta's state-of-the-art interactive segmentation tool, the Segment Anything Model (SAM) (Kirillov et al. 2023), which is pre-trained on a massive and diverse dataset of over 1 billion masks. SAM's generalization and interactivity (allowing user-placed markers that guide the model) might thus allow for semi-automatic annotation of Linear A inscriptions without the need for large amounts of training data, which is unfortunately not available in this context.[9]

[9] By the "interactivity" of SAM, what is meant is NOT any built-in user interface in the form of a GUI, but rather the possibility of prompt-based placement of rectangles of focus, and points of additions and deletions that guide the model's prediction.

1.2 Prior work

This thesis places itself at the intersection of computer vision, digital humanities, and epigraphy (the study of inscriptions). Since it is a thesis of engineering, computer vision will remain the primary focus, both in terms of the research questions, and the methods used to answer them. As a result, the other two fields will merely provide the context and motivation for the research. The prior work and initial literature review in focus will thus concentrate on: 1. the relevant theory and state-of-the-art within computer vision (SAM, etc.), 2. the palæographical and digital humanities work that this project will seek to support (SigLA), and 3. the overall context thereof (Linear A).

1.2.1 Research context

For the context of Linear A and its corpus, the background section above can be referred to, citing the sources for: its size (Salgarella 2020) and (Salgarella 2025), its new addition (Del Freo and Zurbach 2025), and where it's accessed (Godart and Olivier 1976–1985). And for the epigraphical and digital humanities work, the background section can also be referred to, citing the sources for SigLA: its paper (Salgarella and Castellan 2020), its about page (SigLA 2021a), and its document repository (SigLA 2021b). Here, additional context can also be provided by Ester Salgarella herself, as she is a co-supervisor of this project, and the main person behind SigLA.

1.2.2 Computer vision literature

As for the relevant theory and state-of-the-art within computer vision, the literature there can be further split into three parts; 1. sources of general image processing, 2. sources concerning SAM and its implementation, and 3. related computer vision script segmentation work for comparison.

For image processing, a general theory book (Gonzalez and Woods 2018) can be cited, as well as more method-specific papers (when other methods are used) like Otsu thresholding (Otsu 1979), and of course the

OpenCV Python documentation (OpenCV Team 2024) for short implementation reference. Then, for SAM, these are: its paper (Kirillov et al. 2023), its lighter implementation MobileSAM (C. Zhang et al. 2023) - and the newest version of that one MobileSAMv2 (Zhao et al. 2023). Lastly, for related segmentation work there exists a paper using deep learning (in contrast to the methodology of this thesis) obtaining results better than state-of-the-art (P. Zhang, Li, and Sun 2024), and a lighter EU-funded historical document segmentation tool *dhSegment* (Oliveira, Seguin, and Kaplan 2018) also using deep learning.

1.3 Problem statement

As previously stated, the aim of this project is to investigate whether parts of the process of creating annotated drawings for SigLA can be automated. More specifically, whether some kind of semi-automatic segmentation tool using interactive and generalized computer vision - specifically the Segment Anything Model (SAM) - can accelerate the workflow, while maintaining high-quality outputs that are still compatible with SigLA's database. But what would be the relative segmentation quality of such a tool? How automatic would it be exactly? And by how much could it improve the efficiency of the workflow? For a decomposition of the problem statement into its three research questions - each accompanied by their respective operationalization - see the following subsection:

1.3.1 Research questions

1. **Segmentation performance**: How well does a SAM-based segmentation approach perform quantitatively in terms of segmentation accuracy, compared to simple classical image processing baselines when applied to Linear A document images? [Operationalization: SAM-based masks will be compared to classical image-processing baselines using overlap metrics (e.g. IoU/Dice) against the ground truth segmentations derived from SigLA's Krita files for a minimum of 25 documents.]

2. **Degree of automation**: To which extent can user interaction be reduced by a SAM-based semi-automatic segmentation workflow, while still maintaining high-quality outputs? [Operationalization: The number of user interactions (e.g. clicks) required to achieve a certain level of segmentation quality (e.g. IoU/Dice) will be measured against ground truth (SigLA's Krita files) for a minimum of 25 documents.]

3. **Workflow efficiency**: How much reduction in annotation time can be achieved using the proposed tool compared to fully manual digitization, under controlled experimental conditions? [Operationalization: The time it takes Ester Salgarella to annotate manually compared to being assisted by the proposed segmentation tool will be measured for a minimum of 5 documents.]

1.4 Empirical considerations

For the evaluation of the solution, a dataset will be compiled together with Ester Salgarella of at least 25 images of original documents from the corpus, selected and cropped from Cefael. These images are subject to Intellectual Property Rights (IPR) issues, and can thus not be displayed publicly if not through Cefael. However, they can still be used for the development and evaluation of the tool (and can be bought if necessary for show-casing within the thesis). For direct reference to the primary source of data, the specific links to the Cefael online webpages for each document used might have to be cited.

As ground truth, their corresponding digitized layered drawings, stored as Krita (.kra) files on SigLA (provided by Ester Salgarella) will be used and aligned for comparison with tool outputs. The ground truth data might then have to be cleared of any marked "deleted signs" or other annotation elements that our model is not intended to reproduce.

The solution will then be built as a pipeline, consisting of pre-processing, semi-automatic segmentation, post-processing, and export (e.g. as Krita files). At last, the primary deliverable of this project will be delivered to the external partner and co-supervisor Ester Salgarella in the form of a proof-of-concept semi-automatic annotation tool. All supporting components, including evaluation results, and analysis will be documented and presented in the written thesis.

1.5 Impact – innovation and application

The work of this thesis will present itself as a first proof-of-concept for the use of modern computer vision to accelerate the expansion of the SigLA database, supporting the further digitization of the Linear A corpus. Its main contribution will be a semi-automatic annotation tool, which might prove useful to epigraphers for generating segmentations of original document images, as they can then be completed and refined more easily. This approach could potentially also be applied to Linear B, as it is the whole field of Aegean studies that seem to suffer from severe under-digitization of the primary evidence, which is a huge problem for the scientific investigations of scholars. Hopefully, it will inspire further research and development within the field of digital epigraphy, particularly in the application of generalized and interactive computer vision approaches such as SAM on Linear A, and perhaps even in the broader context of use on small corpus ancient writing systems where the lack of big data is ill fit for approaches using deep learning.

[TODO: FIX THE INTRO AND FIT IT TO THE THESIS]

1.6 Overview of the thesis

The rest of this thesis is structured into five main chapters (Methodology, Data, Results, Discussion, Conclusion), all split up into sections concerning each of the three research questions.

CHAPTER 2

Data

2.1 Sources

The dataset used in this thesis was constructed from photographs of raw Linear A inscriptions, and their corresponding ground truth annotations. Cefael (see Godart and Olivier 1976–1985) was chosen as the source for raw document images, since it is the only online and widely publicly available source of the printed five volume corpus - Godart and Olivier, Recueil des inscriptions en linéaire A (GORILA). As all raw document images on the pages of this corpus are accompanied by annotated drawings, an argument could be made to make use of those as ground truth annotations. However, annotations provided by Ester Salgarella from SigLA (see Salgarella and Castellan 2020) were chosen instead, as they are more up to date, and are already in a binary mask format that is suitable for comparison with segmentation model outputs.

What follows in this chapter is a detailed description of this dataset, documenting its characteristics, its construction process, and its limitations and biases.

2.1.1 Characteristics and construction

The dataset was made to consists of 32 chosen image pairs of raw documents and their corresponding ground truth annotations. Only inscriptions with clay as writing support were included, as they are the most common and representative of the Linear A corpus, and are the only ones for which SigLA ground truth annotations were already available (IS THIS TRUE?). The raw document images from Cefael are screenshots of images within the scanned printed edition of the corpus, and thus are not of the highest quality, but they are the only widely publicly available images of the Linear A inscriptions.

((a)) Raw image (HT7b) **((b))** Ground truth (HT7b)

Figure 2.1: Example of a raw inscription image and its corresponding ground truth annotation.

2.1.2 Difficulty classification

The dataset was categorized into three difficulty levels (easy, medium, hard) based on the visual quality of the raw images and the clarity of the inscriptions. This classification was performed manually by inspecting each image and considering factors such as contrast, noise, and the presence of tablet breakages or general damage to the inscriptions. Concretely, the following criteria were used to classify each difficulty set:

X

- Easy: labeled as such for documents with clear ingravings, good contrast, minimal noise, and constant lighting.
- Hard: labeled to those with poor contrast, significant noise, and substantial damage making it hard even for the naked eye to read the inscriptions.
- Medium: labeled to all those inbetween the qualities of easy and hard.

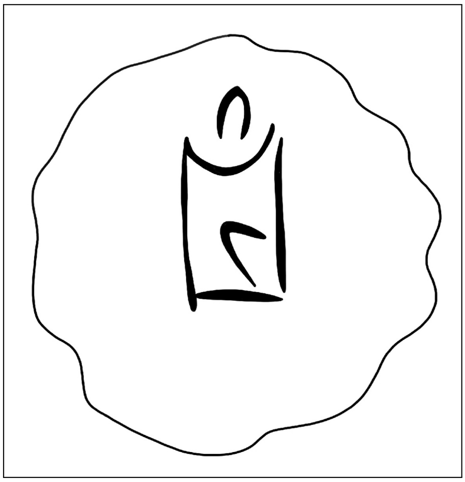
Figure 2.2: Criteria for classifying the dataset into easy, medium, and hard difficulty levels.

2.1.3 Image registration and preprocessing

To obtain any kind of meaningful comparison between model outputs and ground truth, the raw images and their ground truth annotations first had to be aligned and registered to each other. This was done manually in Krita by downscaling the ground truth images to the individual scales of their raw counterparts, and aligning by translation and rotation, using the raw images as background reference.

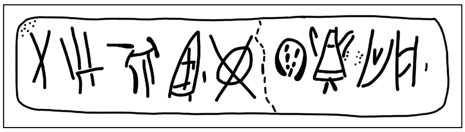
2.2 Limitations and biases

((a)) Easy KHWc2104 (raw)

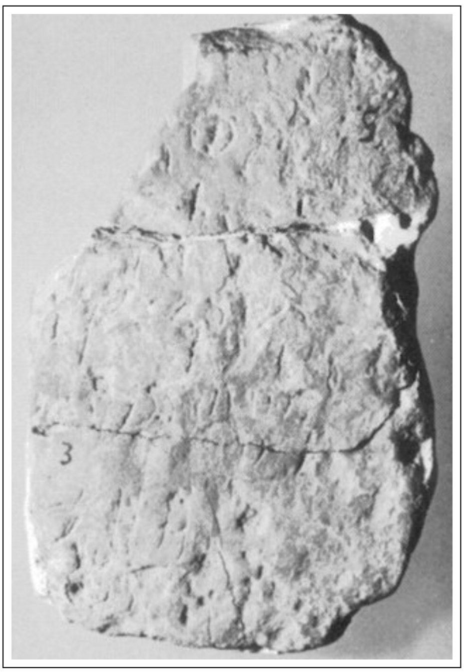


((b)) Easy KHWc2104 (gt)

((c)) Medium MA1a (raw)



((d)) Medium MA1a (gt)



((e)) Hard HT3 (raw)

((f)) Hard HT3 (ground truth)

Figure 2.3: Examples of images with different levels of segmentation difficulty: easy, medium, and hard. For each case, the raw image (left) and the corresponding ground truth annotation (right) are shown.

CHAPTER 3

Methodology

3.1 Common methodology

[TODO: WRITE A C]

3.1.1 Image processing theory

[SHOULD THIS SECTION EXIST? IF SO, I SHOULD FINISH IT]

The basics of spatial filtering can be described by the following equation:

$$g(x, y) = T[f(x, y)]$$

, where $f(x, y)$ (also denoted as $I(x, y)$) is the input image, $g(x, y)$ is the output image.[1]

3.1.1.1 Binary image thresholding

$$B(x, y) = \begin{cases} 1 & \text{if } I(x, y) > T \\ 0 & \text{otherwise} \end{cases}$$

[2]

[1]Gonzalez and Woods 2018, p. 120.
[2]Gonzalez and Woods 2018, p. 743.

3.1.1.2 Convolutional kernels

[3]

$$(w \star I)(x,y) = \sum_{s=-a}^{a} \sum_{t=-b}^{b} w(i,j) I(x-i, y-j)$$

where w is the convolution kernel (window) of size $m \times n$, and $a = (m-1)/2$ and $b = (n-1)/2$ are the half-width and half-height (assuming odd dimensions), respectively.[4]

3.1.2 Development environment and tools

[TODO: EXTEND THIS SECTION] Python 3.10 was used for all code development (because it has such a big library of image processing functions and machine learning model implementations), with the following libraries:

- OpenCV for image processing and computer vision tasks.
- NumPy for numerical operations and array manipulations.
- Matplotlib for data visualization.
- *Add more!*

Krita was used for manual image registration, making sure the raw image ground truth pairs were properly aligned.

3.1.3 General evaluation metrics

3.1.4 Experimental conventions

3.2 Segmentation quality

Here, the one-shot segmentation quality of SAM will be evaluated against a set of baseline models, using a variety of performance metrics.

[3] Gonzalez and Woods 2018, p. 159.
[4] Gonzalez and Woods 2018, p. 157.

[Add some context of RQ1 here, and how the question will be operationalized in bigger picture (though more detail than in introduction).]

3.2.1 Baseline model configuration

The baseline models against which the tool will be evaluated (from simplest to most advanced) are:

- Global thresholding (Otsu's method)
- Local adaptive thresholding (Gaussian kernel)
- Edge-based segmentation (Canny edge detection with region filling)
- Energy-based segmentation (GrabCut)

Together, they cover a wide range of classic segmentation techniques that might be used for binary foreground extraction. What follows is a brief description of each of these methods, including their characteristics, how they work, and their implementation details. [TODO: ADD MORE IMPLEMENTATION DETAILS BESIDES PURE THEORY] [SHOULD I REDUCE THEORY? HOW LONG SHOULD THESE BE? PROBABLY NOT MORE THAN A PAGE FOR EACH BASELINE]

3.2.1.1 Global thresholding (Otsu's method)

This is a global thresholding method from gray-level histograms originally proposed by Otsu (1979). As the simplest of the baselines, it represents the minimum acceptable segmentation quality, since it only takes the global pixel intensity into account, with no assumptions about any spatial relationships between pixels. It might therefore perform poorly on images with erosion, uneven lighting, and differences in surface texture[5], which are the exact characteristics of the Linear A inscriptions to be segmented.

Here is how it works: First, let pixels of the raw image be divided into L shades of gray $[1, 2, ..., L]$. Then, let each pixel be assigned one of two classes C_0 or C_1, guided by a threshold T, where C_0 contains the shades

[5] Gonzalez and Woods 2018, p. 751.

$[1, 2, ..., T]$, and C_1 contains the rest $[T+1, T+2, ..., L]$. Otsu's method then works by performing an exhaustive search over all L gray levels to find an optimal threshold T^* that results in a maximum between-class variance σ_B^2:

$$T^* = \underset{1 \leq T < L}{\arg\max}\ \sigma_B^2(T)$$

The between-class variance σ_B^2 can then be defined by a function of the respective total probabilities of the two classes (ω_0 and ω_1) and the mean gray level value per class (μ_0 and μ_1):

$$\sigma_B^2(T) = \omega_0(T)\omega_1(T)[\mu_1(T) - \mu_0(T)]^2$$

For the implementation of Otsu's method in OpenCV, the following parameters are used:

- `cv2.THRESH_OTSU` for the thresholding type, and
- `cv2.THRESH_BINARY` for the binary thresholding method.

3.2.1.2 Local adaptive thresholding (Gaussian kernel)

Using local adaptive thresholding is an advantage in cases where the image has varying light conditions (which is the case for the Linear A inscriptions), as the filter adapts to local lighting. This is due to the fact that the computed threshold T is not global, but instead computed for each pixel based on its local neighborhood.

Mathematically, T is a function of a given pixel position (x, y) usually computed by some kind of weighted sum of the neighborhood. The generated binary mask B is then defined as:[6]

$$B(x,y) = \begin{cases} 1 & \text{if } I(x,y) > T(x,y) \\ 0 & \text{otherwise} \end{cases}$$

To compute $T(x, y)$, there are a few weighting functions to choose from. But for this baseline, the Gaussian kernel is used, as it tends to be better at handling noise, since it assigns weights based on distance

[6] Gonzalez and Woods 2018, p. 761.

from the kernel center. The pixels closer to the center will have a higher weight, while those farther away will have a lower weight, which limits the influence of distant noisy pixels.

With a Gaussian kernel G_σ, the threshold $T(x, y)$ is computed as a Gaussian-weighted sum of local intensities subtracted by a constant C. Using convolution ($\star$), this can be expressed as:[7]

$$T(x, y) = (G_\sigma \star I)(x, y) - C$$

The weight at a given pixel offset (i, j) from the kernel center (where α is a normalization constant ensuring that the sum of the kernel elements equals 1, and σ is the standard deviation of the weights) is then defined as follows:[8]

$$G_\sigma(i, j) = \alpha \cdot \exp\left(-\frac{i^2 + j^2}{2\sigma^2}\right)$$

The OpenCV implementation[9,10] uses the following parameters:

- `cv2.ADAPTIVE_THRESH_GAUSSIAN_C` for the thresholding type, and
- `cv2.THRESH_BINARY` for the binary thresholding method.

3.2.1.3 Edge-based segmentation (Canny edge detection with region filling)

This is a method that by itself only segments the edges of an image, excluding the enclosed regions within those edges. Here is how it works: First, the input image is smoothed using a Gaussian filter to reduce noise (using the Gaussian kernel as defined in the previous section):

$$I_s(x, y) = (G_\sigma \star I)(x, y)$$

[7] OpenCV 2024a.

[8] Gonzalez and Woods 2018, p. 167.

[9] OpenCV 2024b.

[10] The Gaussian kernel is *separable* and *isotropic*, meaning that it can be implemented (like done in OpenCV) as two consecutive 1-D convolutions, reducing computational complexity from $O(n^2)$ to $O(n)$ for a kernel of size $n \times n$. (Gonzalez and Woods 2018, p. 167).

Then, for the image gradient of each pixel, the magnitude $M_s(x, y)$ and the orientation $\theta_s(x, y)$ are computed:[11]

$$M_s(x, y) = \| \nabla I_s(x, y)\| = \sqrt{\left(\frac{\partial I_s}{\partial x}\right)^2 + \left(\frac{\partial I_s}{\partial y}\right)^2}$$

$$\theta_s(x, y) = \arctan\left(\frac{\partial I_s/\partial y}{\partial I_s/\partial x}\right)$$

Next, *non-maxima suppression*[12] is applied to retain only local maxima of $M_s(x, y)$ along the gradient direction $\theta_s(x, y)$, resulting in thinner edges. This is at last followed by *hysteresis thresholding*[13], classifying pixels as strong, weak, or non-edges with a low threshold T_l and a high threshold T_h, only preserving weak edges if connected to strong edges.
[Add details about region filling here.]
The OpenCV implementation[14,15] uses the following parameters:

- `cv2.Canny` for the edge detection method.

3.2.1.4 Energy-based segmentation (GrabCut)

GrabCut - the most advanced of the baselines considered - is included, as it is an interactive segmentation method, which makes for a good comparison to SAM. It is also good at producing spatially coherent masks with little large-grained noise, which is useful on Linear A inscriptions, where erosion usually leads to noisy masks even after noise reduction. Here is how it works: For energy-based segmentation methods, optimal segmentation is achieved by minimizing a cost (energy) function E that combines a data term D and a smoothness term S, given an input image I, a binary segmentation mask B, and a set of parameters θ:[16]

[11] Gonzalez and Woods 2018, p. 730.
[12] Gonzalez and Woods 2018, p. 730-731.
[13] Gonzalez and Woods 2018, p. 732.
[14] OpenCV 2024b.
[15] The Gaussian kernel is *separable* and *isotropic*, meaning that it can be implemented (like done in OpenCV) as two consecutive 1-D convolutions, reducing computational complexity from $O(n^2)$ to $O(n)$ for a kernel of size $n \times n$. (Gonzalez and Woods 2018, p. 167).
[16] This is a simplified version of the energy function that is used by GrabCut.

$$E(B, I, \theta) = D(B, I, \theta) + S(B, I)$$

In the case of GrabCut (Rother, Kolmogorov, and Blake 2004), the algorithm is interactively initialized by the user, who provides a bounding box around the object of interest. Pixels outside this box are marked as definite background, while pixels inside it are marked as initially unknown. To further guide the segmentation, the user also has the option to provide an additional number of pixels as definite foreground ($B_n = 1$), and definite background ($B_n = 0$).

The set of parameters θ then define two *Gaussian Mixture Models (GMMs)* (probabilistic models each consisting of a weighted sum of several Gaussian distributions) that over iterations will be tuned to model the foreground and background pixel distributions. These are used to compute the data term D whose energy contribution decreases the better B matches the GMMs of foreground and background. The smoothness term S then penalizes neighboring pixels that have different labels, encouraging spatial coherence in the segmentation.

At last, the optimal segmentation B^* is found iteratively, by alternating between optimizing the GMM parameters θ to model B, and minimizing the energy function with respect to B:

$$B^* = \arg\min_{B} E(B, I, \theta)$$

[THIS CLASSICAL INTERACTIVE SEGMENTATION METHOD MIGHT FOLLOW SAM INTO RQ2]

[MAYBE IT SHOULD BE GIVEN MORE IMPORTANCE?]

3.2.2 State-of-the-art model configuration

3.2.2.1 Model architecture

Raw image → Automatic prompts → MobileSAMv2 → Binary mask

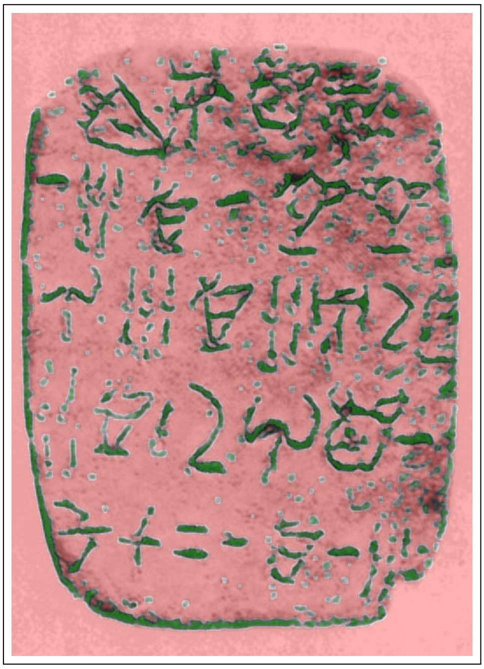


Figure 3.1: A visualization of seed distribution for GrabCut with brush seeding (GC+brush) on HT118. Green overlay represents sure foreground pixels, while red overlay represents sure background pixels, and absence of overlay represents probable background pixels.

3.2.2.2 Segmentation (MobileSAMv2)

For the sake of computational efficiency, a mobile implementation of SAM - MobileSAMv2 (mSAM) (see Zhao et al. 2023) will be used instead of the full SAM model. This is a lightweight version of the original SAM model, which is designed to run efficiently on mobile devices while still maintaining good segmentation performance. It is ... but trained on the same dataset as the original SAM, and thus has the same one-shot segmentation capabilities, although with a slightly lower performance than the original.

Here is how it works: ViT encoder, prompt encoder, mask decoder, etc.

3.2.2.3 Seed extraction

Even though the mSAM implementation already makes use of an automatic box prompt, it does so from a YOLO model trained on the same dataset as SAM. Those box prompts are generalized for and work best on natural images. Hence, they may not perform well on gray-scale images of clay tablets with Linear A inscriptions, as those were not part of the training data. Thus, mSAM will need to be adapted through the use of prompts. The model takes two types of prompts: 1. a single box prompt, and/or 2. point prompts labeled as either foreground or background. For the one-shot segmentation capability of the model, these prompts will have to be automatically generated, rather than interactively placed by a user. However, as it is particularly difficult to automatically create high quality box prompts without a trained object detection model, only point prompts will be used. Therefore, there is to be made two versions of MobileSAMv2 for comparison:

- **mSAM**: MobileSAMv2 (with the original automatic box prompts from the YOLO model),

- and **mSAM+pts**: MobileSAMv2 with automatic point prompts labeled as either foreground or background.

The automatic point prompts for mSAM+pts (see Figure 3.2 for visualization) will be derived from the thresholded mask of the Gaussian adaptive thresholding baseline (see Section 3.2.1.2), as it seems to be the best at extracting foreground from the Linear A tablets, albeit with a degree of noise. This will be done by randomly sampling a number of foreground and background pixels from the mask as seeds for mSAM+pts. For the sake of simplicity, this number will be kept equal for both point types to a 1000 points each, which should be large enough to provide good coverage of the image, while still being computationally feasible for the model to process.

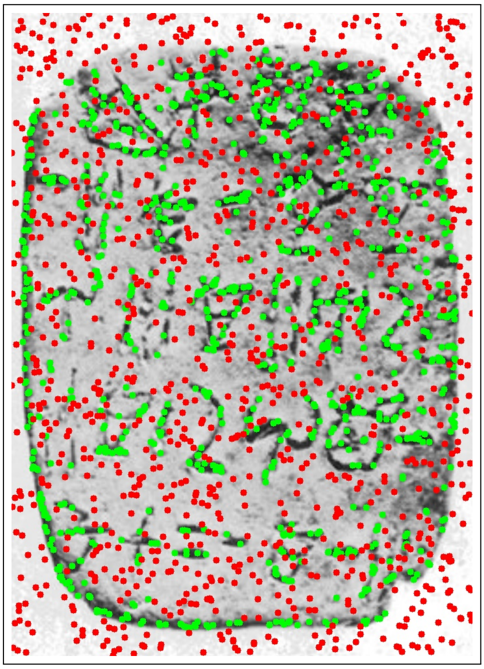


Figure 3.2: A visualization of seed distribution for MobileSAMv2 with automatic point prompts (mSAM+pts), overlayed on HT118. Green points represent foreground seeds, while red points represent background seeds.

3.2.3 Evaluation protocol and performance metrics

For an evaluation of one-shot segmentation quality, relevant metrics and statistical tests will need to be defined to be able to determine whether one model performs significantly better than another. More specifically, it is of interest to compare the best performing baseline model against the best performing version of MobileSAMv2.

The following subsections will therefore document the choice of metrics and statistical tests for the evaluation, used to answer RQ1: *How does the one-shot segmentation quality of SAM compare to that of classic segmentation methods?*.

3.2.3.1 Choice of metrics

When it comes to evaluation of segmentation models, the two most commonly used metrics are: 1. the Dice-Sørensen coefficient (Dice) and 2. the Intersection over Union (IoU). The Dice score is most commonly used in cases where there are heavy class imbalances (e.g. lots of background compared to regions segmented), and when foreground objects are small and thin. It is naturally more foregiving than the related segmentation metric IoU, which instead tends to be rather strict, and is specifically used to evaluate segmentation results where small displacements are critical for results. Therefore, this thesis will report segmentation performance using Dice. Mathematically, it is defined as twice the size of the intersection of the predicted segmentation mask $\hat{Y}$ and the ground truth mask Y, divided by their summed size:[17]

$$\mathrm{Dice}(\hat{Y}, Y) = \frac{2 \cdot |\hat{Y} \cap Y|}{|\hat{Y}| + |Y|}$$

Thus, the raw metrics will end up consisting of the Dice score for each image, and each model.

3.2.3.2 Statistical tests

To determine whether the observed differences in scores between models are statistically significant, paired t-tests will be needed. The paired t-test has two main assumptions: 1. independence of observations, and 2. normality of residuals. The first assumption is already satisfied, since the segmentation performance of one image does not directly influence the performance on another. However, the second assumption depends on the results. Thus, the residuals of the paired differences in Dice scores between the two models in question will be checked for normality using a QQ-plot and the Shapiro-Wilk test. And in case the normality assumption is violated, the non-parametric domain-specific Wilcoxon signed-rank test will be used instead.[18]

[17] Mohammadi, Mollazade, and Behroozi-Khazaei 2025.
[18] Rainio, Teuho, and Klén 2024.

3.3 Interactive semi-automation

Here, the interactivity and quality of human-guided SAM (and GrabCut for comparison?) will be evaluated against the one-shot results from Section 3.2. [Add some context of RQ2 here, and how the question will be operationalized in bigger picture (though more detail than in introduction).]

3.3.1 Interactive tool architecture

3.3.2 Experimental design for automation evaluation

3.3.3 Definition of automation metrics

3.4 Workflow efficiency

[Add some context of RQ3 here, and how the question will be operationalized in bigger picture (though more detail than in introduction).]

3.4.1 Definition of workflow efficiency metrics

3.4.2 Experimental design of the user study

3.4.3 Analysis of results

CHAPTER 4

Results

4.1 Segmentation performance

4.1.1 Statistical significance

4.1.1.1 Normality of residuals

4.1.1.2 Paired t-test

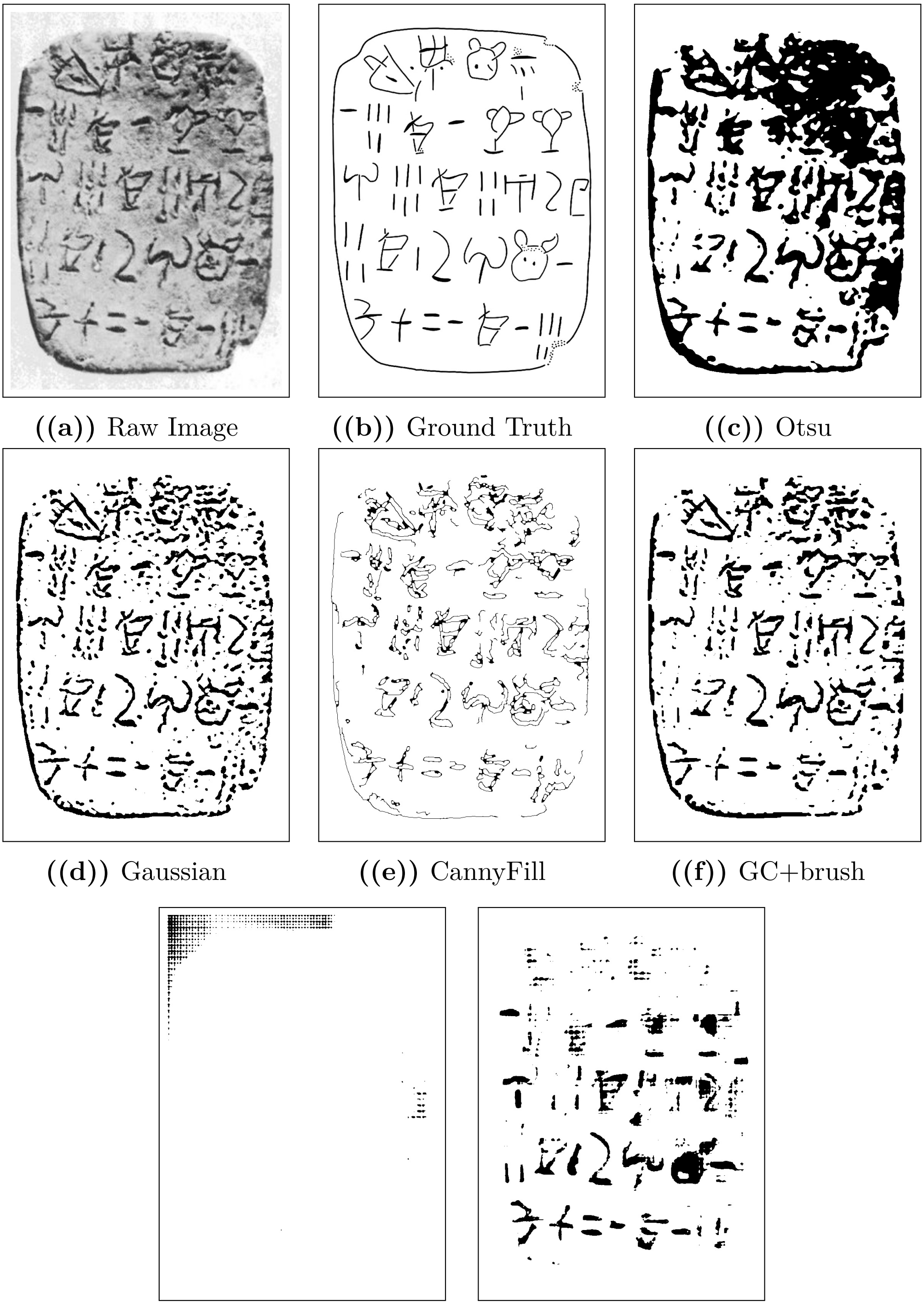
To determine whether the classic baseline model GC+brush signficiantly outperforms mSAM+pts, a paired t-test is conducted on the Dice scores obtained for each image across these two models. The null hypothesis (H_0) states that there is no difference in mean Dice scores between the two models, while the alternative hypothesis (H_1) states that GC+brush has a higher mean Dice score than mSAM+pts.

$$\delta = \mu_1 - \mu_2 = 0.18 - 0.14 = 0.04$$

$$t = 5.22$$

$$P(T > t) = 1.25 \cdot 10^{-5}$$

Thus, as $1.25 \cdot 10^{-5} < \alpha = 0.05$, the null hypothesis can be rejected, and it can be concluded that GC+brush significantly outperforms mSAM+pts in terms of Dice score. Hence, it can be said that mSAM underperforms in comparison to classical segmentation methods for one-shot segmentation of Linear A tablets.



((a)) Raw Image ((b)) Ground Truth ((c)) Otsu

((d)) Gaussian ((e)) CannyFill ((f)) GC+brush

((g)) mSAM ((h)) mSAM+pts

Figure 4.1: Comparison of segmentation outputs from different models against the raw image and ground truth.

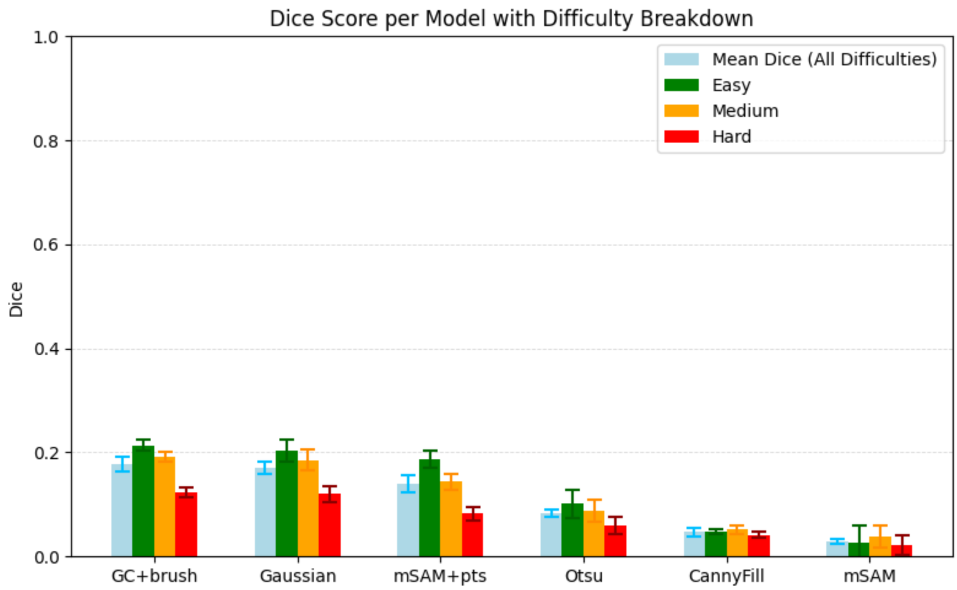


Figure 4.2: Dice Score performance per model. Split by difficulty. With error bars representing the standard error of the mean. Ordered descendingly from the left by mean performance.

4.2 Automation evaluation

4.3 Workflow efficiency evaluation

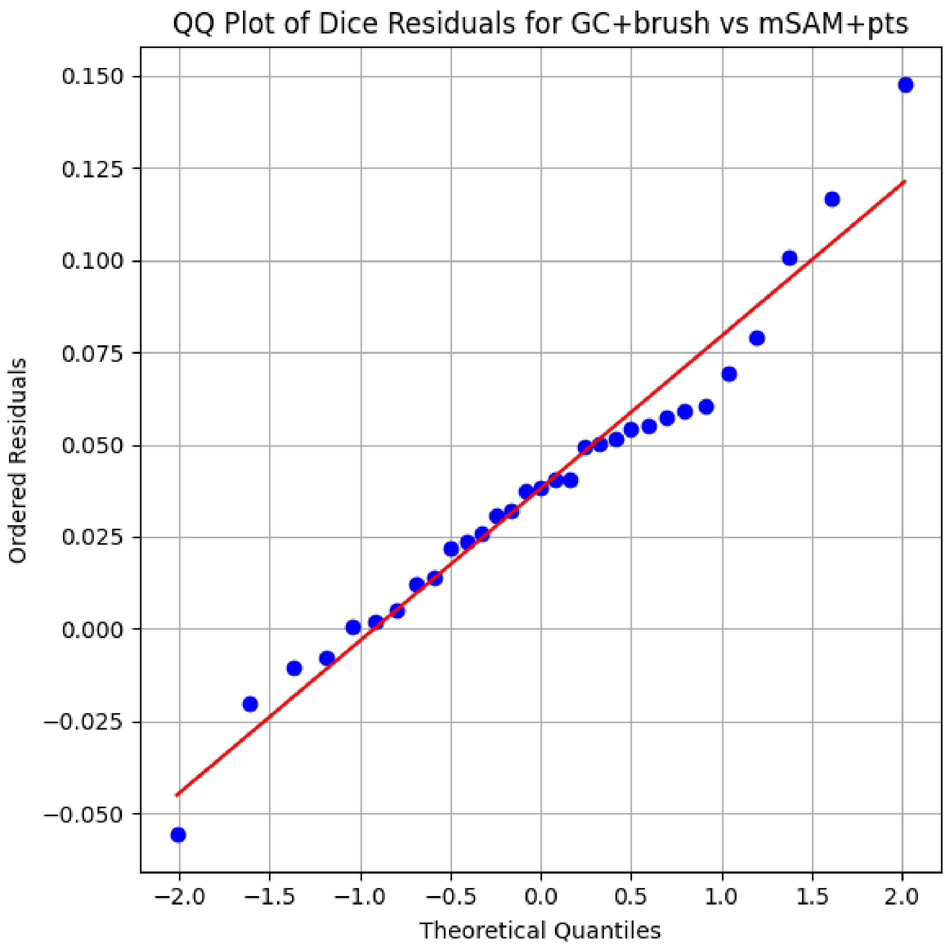


Figure 4.3: QQplot of the residuals of the paired t-test comparing G+brush and mSAM+pts. The residuals appear to be approximately normally distributed, which supports the validity of the paired t-test results. Furthermore, the Shapiro-Wilk test for normality on these residuals yields a p-value of $5.9 \cdot 10^{-1} > \alpha = 0.05$, indicating that we fail to reject the null hypothesis of normality.

CHAPTER 5

Discussion

5.1 Segmentation ability

Florence 2 (Wang et al. 2025)

5.2 Automation and interactivity

5.3 Workflow impact

CHAPTER 6

Conclusion

6.1 Summary of findings per research question

6.1.1 RQ1: Segmentation performance

6.1.2 RQ2: Degree of automation

6.1.3 RQ3: Workflow efficiency

6.2 Overall contribution

6.3 Final remarks

CHAPTER 7

Future Work

7.1 Methodological extensions

7.2 Dataset expansion and diversification

7.3 Further automation opportunities

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APPENDIX A
An Appendix

A.1 Raw segmentation performance

Table A.1: Raw Dice score data on easy images. Label column represents the name of the document segmented, while all other columns represent Dice scores obtained for those documents for a given model.

Label	CannyFill	GC+brush	Gaussian	Otsu	mSAM	mSAM+pts
HT118	0.075	0.169	0.162	0.1	0.002	0.137
HT12	0.039	0.26	0.259	0.176	0.048	0.199
HT154B	0	0.143	0.147	0.07	0.032	0.088
HT17	0	0.23	0.215	0.072	0.03	0.251
HT22	0	0.192	0.187	0.13	0.057	0.161
HT26a	0.029	0.263	0.256	0.11	0.001	0.205
HT40	0.116	0.161	0.161	0.074	0.029	0.111
HT7a	0.111	0.129	0.136	0.109	0.02	0.124
HT7b	0.15	0.18	0.162	0.073	0.026	0.123
HT90	0	0.179	0.181	0.131	0.012	0.154
KHWc2104	0	0.442	0.37	0.067	0.027	0.497

Table A.2: Raw Dice score data on medium images. Label column represents the name of the document segmented, while all other columns represent Dice scores obtained for those documents for a given model.

Label	CannyFill	GC+brush	Gaussian	Otsu	mSAM	mSAM+pts
HT10a	0.037	0.254	0.249	0.119	0.057	0.154
HT16	0.022	0.177	0.159	0.08	0.003	0.165
HT2	0.013	0.114	0.127	0.082	0.021	0.076
HT4	0.007	0.237	0.239	0.097	0.061	0.213
HT6b	0.151	0.188	0.175	0.071	0.004	0.147
HT85b	0.056	0.118	0.106	0.058	0.035	0.081
HT8b	0.101	0.274	0.267	0.092	0.052	0.157
HT94b	0.059	0.083	0.085	0.035	0.017	0.032
MA1a	0.066	0.232	0.227	0.146	0.09	0.243
PH2	0.003	0.242	0.224	0.095	0.046	0.172

Table A.3: Raw Dice score data on hard images. Label column represents the name of the document segmented, while all other columns represent Dice scores obtained for those documents for a given model.

Label	CannyFill	GC+brush	Gaussian	Otsu	mSAM	mSAM+pts
HT1	0.001	0.117	0.103	0.045	0.049	0.068
HT27a	0.093	0.153	0.148	0.056	0.047	0.005
HT3	0.031	0.08	0.089	0.054	0.046	0.078
HT30	0.065	0.096	0.098	0.05	0.006	0.082
HT33	0.043	0.054	0.051	0.016	0	0.033
HT52a	0	0.1	0.109	0.059	0.006	0.06
HT85a	0.102	0.084	0.097	0.055	0.006	0.092
HT8a	0.08	0.206	0.186	0.056	0.041	0.127
HT9a	0.002	0.219	0.212	0.137	0.02	0.219
HT9b	0.001	0.117	0.115	0.067	0	0.063

Table A.4: Summary Statistics per Model of Dice Score on all images.

Model	Mean Dice	Standard deviation	Standard error	n
CannyFill	0.047	0.047	0.008	31
GC+brush	0.177	0.08	0.014	31
Gaussian	0.171	0.069	0.012	31
Otsu	0.083	0.036	0.006	31
mSAM	0.029	0.023	0.004	31
mSAM+pts	0.139	0.092	0.016	31

Table A.5: Summary Statistics per Model of Dice Score on easy images.

Model	Mean Dice	Standard deviation	Standard error	n
CannyFill	0.047	0.056	0.017	11
GC+brush	0.214	0.087	0.026	11
Gaussian	0.203	0.069	0.021	11
Otsu	0.101	0.035	0.01	11
mSAM	0.026	0.017	0.005	11
mSAM+pts	0.186	0.113	0.034	11

Table A.6: Summary Statistics per Model of Dice Score on medium images.

Model	Mean Dice	Standard deviation	Standard error	n
CannyFill	0.052	0.047	0.015	10
GC+brush	0.192	0.067	0.021	10
Gaussian	0.186	0.064	0.02	10
Otsu	0.088	0.031	0.01	10
mSAM	0.039	0.028	0.009	10
mSAM+pts	0.144	0.064	0.02	10

Table A.7: Summary Statistics per Model of Dice Score on hard images.

Model	Mean Dice	Standard deviation	Standard error	n
CannyFill	0.042	0.041	0.013	10
GC+brush	0.123	0.054	0.017	10
Gaussian	0.121	0.048	0.015	10
Otsu	0.06	0.031	0.01	10
mSAM	0.022	0.021	0.007	10
mSAM+pts	0.083	0.058	0.018	10

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

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